LINAS BERESNA

Vancouver, Canada
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EDUCATION

PhD in Computer Graphics, Simon Fraser University, Canada *2021 - Present*
Supervised by Eugene Fiume.
MEng Computer Science & Mathematics, University College London, UK *2015 - 2019*
Graduated with first-class honours. Exchange at University of Waterloo, Canada (2017 - 2018).

TECHNICAL SKILLS

Advanced	C++, gdb, Linux, Python, Mitsuba
Intermediate	Git, OSL, USD
Basic Knowledge	Houdini, Maya, Javascript, OpenGL, Qt

EXPERIENCE

Image Engine Design June 2022 - May 2025; Sept 2025 - Present
Part-time Rendering Engineer *Vancouver, Canada*

· Supporting lighting and lookdev departments by maintaining legacy software and Gaffer integration.
· Developing and optimizing rendering tools using C++, Python, and OSL.

Disney Research Zurich May 2025 - August 2025
Research Intern - Digital Humans *Zurich, Switzerland*

· Conducted research and development within the Digital Humans team.

Animal Logic July 2020 - August 2021
RND Rendering Engineer *Sydney, Australia*

· Built and supported the proprietary renderer Glimpse using C++14, USD, and OptiX.
· Created artist-facing tools and shaders for Maya, Houdini, and in-house applications.

DNEG July 2019 - July 2020
RND Software Engineer - On-Set Tools *London, UK*

· Created and supported tools for the shoot department to automate importing and sorting large datasets.
· Developed cross-platform applications using C++, Python, and Qt.

Gambit Research August 2018 - September 2018
Software Engineer Intern *London, UK*

· Redesigned user-input logging services, migrating the codebase from Node.js to Python.
· Utilized Docker and Kubernetes for containerized deployment.

UCL - Surgical Robot Vision Group June 2018 - August 2018
Research Intern *London, UK*

· Investigated binary classification for cancer cells using Deep Learning (TensorFlow).
· Automated microscope hardware using C++ and OpenCV for edge detection and auto-focusing.

PUBLICATIONS

Beresna, L., Fiume, E. (2026). Sensitivity as an Arbitrary Output Variable for Differentiable Rendering. *SIGGRAPH Asia Technical Communications.*

Beresna, L., Fiume, E. (2026). Stochastic Graph Compression for Constant-Memory Differentiable Light Tracing. *SIGGRAPH Asia Technical Communications.*

Beresna, L., Fiume, E. (2026). Scene Parameter Saliency via Differentiable Light Transport. *British Machine Vision Conference (BMVC).*

Beresna, L., Fiume, E. (2026). Glare Mitigation using a Differentiable Unified Glare Rating. *Eurographics Symposium on Rendering – DL-only Track*.

Neary-Zajiczek, L., **Beresna, L.**, et al. (2023). Minimum resolution requirements of digital pathology images for accurate classification. *Medical Image Analysis*, 89:102891.